EMERITUS

Learn. From the world's best.

Paradigms of Machine Learning - II

EMERITUS

Learn. From the world's best.

Why Strategy Matters in ML

Paradigm Impact

Impacts data needs, time, accuracy and trust

Resource Efficiency

Saves time and money with the right choices

Success Rate

Boosts chances of project success

Constraints That Affect ML Decisions

Factors That Shape Machine Learning Choices

Constraint	Examples
Data Volume	50 labelled records only
Label Cost	Expensive expert labeling
Privacy	Cannot send data to server
Compute	Runs on mobile device only
Domain Gap	Training vs. deployment mismatch

Learning Paradigms in AI

Different Ways AI Models Learn From Data

Supervised

Learns using labelled data

Self-Supervised

Generates its own labels from raw data

Transfer Learning

Uses knowledge from pre-trained models

Few-shot / One-shot

Learns from very few examples

Federated

Trains on multiple devices without sharing data

Important Questions to Consider

Factors That Guide the Choice of Learning Approach

Label Availability

Are labels available?

Data Location

Is the data centralised?

Inference Speed

Is real-time output needed?

Transfer Potential

Can we pretrain on other data?

Domain Shift

Will the target domain be different?

Paradigm Trade-offs

Key Factors to Weigh When Choosing A Learning Approach

Label Need
How much labelled data is needed

Data Volume
Total data required for good results

Privacy Fit
Meets privacy requirements

Generalisability
Works well across domains

Compute Need
Processing power needed

Learning Paradigms at a Glance

Quick Comparison of Learning Types, Data Needs and Use Cases

Paradigm	Labels Required	Training Data Size	Adaptability	Used In
Supervised	✅ High	✅ High	❌ Low	Spam filters, fraud detection
Unsupervised	❌ None	✅ High	❌ Low	Clustering, dimensionality reduction
Semi-supervised	⚠️ **Few**	✅ High	⚠️ **Medium**	Sentiment analysis with partial labels
Self-supervised	❌ None	✅ Very High	✅ High	Pre-training for NLP/vision models
Active Learning	✅ Selective	⚠️ **Medium**	✅ Iterative	Medical imaging, rare labelling
Transfer Learning	✅ Pretrained	❌ Low (target)	✅ High	Language models, domain-specific tasks
Few-shot/One-shot	✅ 1–5 samples	❌ Very Low	✅ Extreme	Face ID, low-resource NLP
Federated Learning	✅ Distributed	✅ Localised	✅ Secure	Mobile keyboards, healthcare AI

Decision-Making Steps

Choose the Right Learning Approach Together

Share the Case

Outline needs and limits

Discuss Options

Compare suitable paradigms

Explain the Choice

Share logic and trade-offs

Case 1: Rural Health Assistant

Designing an AI tool For Remote Healthcare Support

Scenario

- Mobile health assistant for rural users
- Categorises symptoms as mild, moderate or severe to guide care

Challenge

- Use lightweight, low-resource models
- Ensure privacy with on-device data handling

Constraints

- Low data
- Strict privacy
- Basic devices

Activity: Choose the Right Approach

Which Method Fits the Challenge Best?

- **Transfer Learning:** Use existing medical models
- **Federated Learning:** Keep data on the device
- **Few-shot Learning:** Train with very few examples

Case 2: Multilingual News Classifier

Detecting Bias and Misinformation in 12 Languages

Scenario

Build a system to classify news by topic and spot bias in 12 languages before elections

Constraints

- Very limited data
- Tight deadlines

Challenge

How to design a language-neutral model that works well with little data and scales quickly?

Activity: Which Approach Fits Best?

Choose The Right Method for Fow-Resource Language Tasks

- **Zero-shot**

  Works without any training examples

- **Few-shot**

  Learns from just a few examples

- **Translate + Supervised**

  Translate to one language, then train normally

Case 3: Classifier for Remote Devices

Smart Image Detection on Low-Power, Offline Devices

Scenario

Use IoT devices (e.g. wildlife or farm cameras) to detect events like pests, crop stages or animals — all offline

Constraints

- Very limited processing power
- No internet access
- 2GB of local, unlabelled images

Learning Challenge

How do we learn from unlabelled images, stay lightweight and support smart local decisions?

Activity: What Would You Choose?

Pick an Approach and Explain Why

Self-supervised

Learn without labelled data

Transfer learning

Start with a pre-trained model

Justify

Explain reasoning

Discuss

Consider trade-offs

Demo Setup: Healthcare Feedback Classifier

Key Challenges

- **Problem:** Classify feedback as positive/negative
- **Few Labels:** Limited labelled examples available
- **Privacy Concern:** Must protect patient data
- **Low Compute:** Restricted processing power

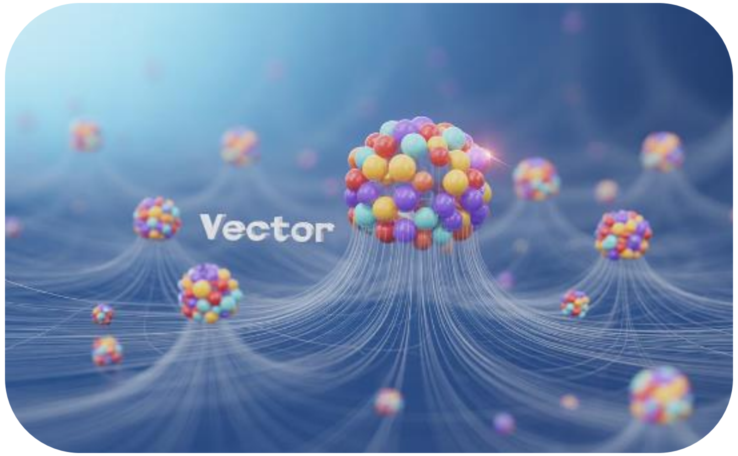
Step 1: Load Pretrained Embeddings

Efficient Model Setup

HuggingFace Integration
Use pre-built models

Embedding Visualisation
Utilise existing knowledge

Implementation
Skip training from scratch



Step 2: Fine-tune on 100 Examples

Rapid Model Adaptation

- **Prepare Data:** Small labelled dataset
- **Fine-Tune Model:** Adapt pre-trained weights
- **Evaluate:** Check performance
- **Iterate:** Quick prototyping cycles

Step 3: Federated Learning Benefits

Privacy-Preserving AI Training

1. **Local Training:** Model trains on individual devices
2. **Data Privacy:** Patient data stays on device
3. **Secure Sharing:** Only model updates are shared
4. **Central Aggregation:** Server combines updates to improve model

Output Review and Discussion

Choosing Your Approach

Centralised Approach

- 87% accuracy | Faster training
- Privacy risks

Federated Approach

- 82% accuracy | Slower convergence
- Robust privacy protection

AI Toolkit Cheat Sheet

Essential Frameworks by Approach

Transfer Learning

HuggingFace | TensorFlow Hub

Few-shot Learning

PyTorch Lightning | OpenAI APIs

Federated Learning

Flower | TensorFlow Federated

Self-supervised

SimCLR | BYOL (PyTorch)

Paradigm-to-Constraint Visual Matrix: Summary

Constraint ↓ \ Paradigm →	Supervised	Transfer	Few-shot	Self-supervised	Federated
Low data availability	❌	✅	✅	⚠️	✅
High label cost	❌	✅	✅	✅	✅
Strong privacy needs	❌	⚠️	✅	✅	✅
Low compute budget	⚠️	✅	✅	⚠️	⚠️

Common Mistakes When Choosing ML Paradigms

Pitfalls of Overusing Few-shot Learning

Be Mindful of When it Works — and When it Doesn't

Few-shot fails with imbalanced or noisy data

Leads to poor results when data isn't evenly distributed

Few-shot isn't a one-size-fits-all

Not ideal for all low-data scenarios

Misusing Transfer Learning

When Pretraining Goes Wrong

Wrong Source: Pretrained model from an unrelated domain

Poor Transfer: Knowledge doesn't apply well

Bad Outcome: Leads to negative transfer

Assuming all pretrained models improve performance without validation

Misunderstanding Federated Learning

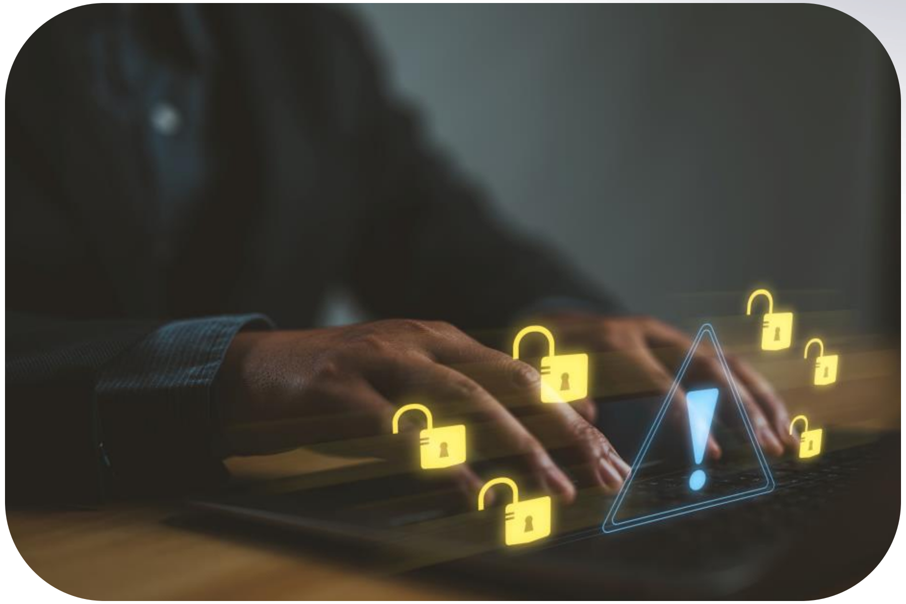
Overlooking Key Risks and Limitations

Security Assumptions:

Not all federated setups are safe from attacks

Performance Limits:

Missed impact of communication and sync delays



Misusing Self-Supervised Learning

Key Pitfalls to Avoid

Undefined Pretext Task: Applying without a clear pretext task

Meaningless Embeddings: Leads to representations that fail to capture useful patterns

Efficiency Assumptions: Assuming it's always more efficient than supervised learning

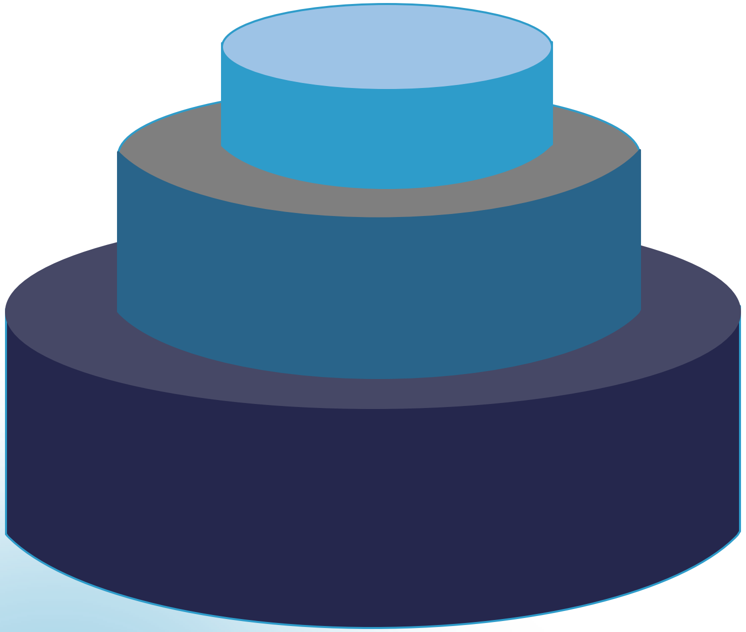
General Pitfall: Ignoring Constraints

Key Oversights to Avoid

- **Data Availability:** Failing to consider data quantity and quality
- **Buzzword-Driven Choices:** Choosing methods based on trends, not suitability
- **Compute Resources:** Overlooking hardware and processing limits
- **Privacy/Legal Boundaries:** Ignoring regulatory and ethical restrictions

Impact of Paradigm Pitfalls

Consequences of Misapplication



- **Project Failure**: Inability to achieve objectives
- **Resource Waste**: Excessive time and budget spent
- **Suboptimal Performance**: Models underperforming against their potential

Misapplying machine learning paradigms leads to these negative outcomes, especially when based on buzzwords rather than data availability, resources and legal/privacy constraints

Best Practices for Paradigm Selection

Making Informed Choices

Thorough Analysis

Evaluate your use case, data, and constraints carefully

Validation Testing

Test assumptions with small experiments before full rollout

Hybrid Approaches

Combine paradigms to balance strengths and weaknesses

Avoiding common pitfalls means focusing on practical needs, not trends. Understand the limits of each paradigm—few-shot, transfer, federated and self-supervised—to choose wisely

Real-World Industry Snapshots

Federated Learning in Action

How Google Gboard Learns Without Compromising Privacy

On-device word prediction

Trains locally on user typing data

Improves suggestions without sharing keystrokes

Preserves user privacy while enhancing performance

Why it Fits

Decentralised data and strict privacy requirements make federated learning ideal

Few-shot / Zero-shot Learning with GPT

Do more with fewer examples

Used for chatbots, tagging, coding help

Works across many tasks

Needs just 1–3 examples

For text generation or classification

Why it works

Fast to deploy, low data needed

Transfer Learning in Medical Imaging

Faster AI Solutions for Healthcare Startups

Use pretrained models

ResNet or EfficientNet on ImageNet

Fine-tune with 500–1000 scans

X-rays or MRIs for medical use

Why it works

Cuts labelling cost and speeds up launch

Self-Supervised Learning with DINO

Powerful Vision Models Without Labelled Data

Vision Transformer without labels:

Advanced computer vision without manual annotation

Learns from unlabelled images:

Extracts meaningful patterns autonomously

For segmentation and anomalies:

Practical applications in various domains

Why it fits:

Unlabelled data is plenty, labels are costly

Domain Adaptation in E-commerce

Adapting Review Models Across Regions

Cross-Region Adaptation

- U.S. models adapted for UK, India, etc.
- Core function kept, regional tweaks applied

Contextual Adjustments

- Adapts tone, product terms and behaviour
- Detects cultural and language nuances

Strategic Advantage

- Reuses core model with minor tweaks
- Efficient for global e-commerce

Q&A Session

Open floor for questions and clarifications.